1. STM32l4S5VI has 3 USARTs and 2 UARTs. Assume LPUART is used for Debug console and not available. How many accelerometers which support UART can be connected? helper (driver) functions to configure NVIC can be found in (there could be more than one correct answer)
    a. 6
    b. 5
    c. 2
    d. 3

2. In a UART with baud rate = 10000; other settings at usual defaults; time to transmit 1 frame is
    a. 1 ms
    b. 10 ms
    c. 0.1 ms
    d. 5 ms

3. Pick the odd one out
    a. COM
    b. tty
    c. Serial
    d. SPI

4. Synchronous protocols generally need only simpler hardware as compared to asynchronous protocols
    a. True
    b. False

5. You are hired by one of the renowned Spacecraft companies as an Embedded Software Engineer to develop the next-generation Guidance, Navigation and Control (GNC) System for future spacecraft. The GNC team has decided to use STM32l475vg as the microcontroller to interface different modules (sensors and actuators) and has provided you with the interface requirements.

    STM32l475vg has 5 UART interfaces (3 USARTs, 2 UARTs) and 3 I2C interfaces.
    Note that USART can operate as UART. Assume LPUART is used for Debug console and not available.

    The following table sums up the interface requirement information for the modules.

Module	Quantity	UART	I2C	
			7-bit address	Speed
IMU	1	115200bps	0x20	100kbps
Sun Sensor	3	-	0b0100xx0	100kbps

			Bits xx: two configurable hardware pins (Designed can use either 0 or 1)	
Star Sensor	1	19200bps	0x20	400kbps
Gyroscope	1	-	0x26	400kbps
Magnetometer	2	19200bps	0x24	400kbps
GPS Receiver	1	115200bps	0x20	100kbps
Reaction Wheel	3	115200bps	-	
Thruster	1	-	0x24	400kbps

(a) List the chosen interface for IMU, Star Sensor, Magnetometer and GPS receiver. Clearly mention the interface number in the form of either UARTx (x = 1 to 5) or I2Cy (y = 1 to 3). Note that both the Magnetometers should either be on I2C or on UART. You are not allowed to have one Magnetometer on I2C and one on UART.

(4 marks)

(b) Based on the answer for (a), what are the two key considerations when you decided to choose (or not choose) I2C interface for IMU, Star Sensor, Magnetometer and GPS receiver?

(2 marks)

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(c) List the chosen I2C addresses of the three Sun Sensors of the GNC system.

(3 marks)

(d) List the speed of the three I2C interfaces for this question.

(3 marks)

(e) Configure:
Pin PC12 for TX signal of UART5 (Use the Alternate function as AF8 for UART5)
Pin PC0 for SCL signal of I2C3 (Use the Alternate function as AF4 for I2C3)
Clearly show all the required register names, their addresses and the configuration values needed for the desired function of the pin. Include a one-line explanation for each configuration step. A program/code is not needed.

(8 marks)